IIT KANPUR
CONSULTING
GROUP

STRATEGY FORGE

PROJECT OBJECTIVES

This project evaluates the feasibility of establishing a mid-sized OSAT (Outsourced Semiconductor Assembly and Testing) facility in India. The analysis goes beyond financial viability and explores:

01 INDIA'S SEMICONDUCTOR ECOSYSTEM AND GROWTH POTENTIAL

02 COMPARISON OF END-USER INDUSTRIES SUCH AS ELECTRIC VEHICLES (EVS), TELECOMMUNICATIONS, CONSUMER ELECTRONICS, AND INDUSTRIAL ELECTRONICS

03 EVALUATION OF POTENTIAL LOCATIONS ACROSS INDIA

04 GOVERNMENT INCENTIVES AND POLICY SUPPORT

05 OPERATIONAL, WORKFORCE, AND SUPPLY-CHAIN READINESS AND LONG-TERM BUSINESS SUSTAINABILITY AND PROFITABILITY

Key Question

Where, why, and how should a semiconductor company expand in India to maximize strategic and financial returns?

PROJECT FRAMEWORK

Market Opportunity Analysis	• Global semiconductor trends • India's semiconductor landscape • Demand drivers across major sectors
Industry Comparison	• Electric Vehicles (EV) • Telecommunications & 5G • Consumer Electronics • Industrial & IoT Applications
Location Assessment	• Gujarat • Tamil Nadu • Karnataka • Other emerging semiconductor hubs
Evaluation Criteria:	• Infrastructure & Talent availability • Logistics connectivity • Supplier ecosystem • Government support
Policy & Incentive Analysis	• India Semiconductor Mission (ISM) • Central government subsidies • State-level incentives
Financial & Business Evaluation	• CapEx and operating costs • Revenue projections • DCF, IRR, NPV & Payback Analysis • Sensitivity analysis
Strategic Recommendation	• Best industry focus • Optimal location choice • Risk assessment • Final expansion blueprint

WHAT IS CONSULTING?

Consulting is the practice of providing expert advice, guidance, or actionable solutions to organizations or individuals in exchange for a fee. The field is generally divided into six core areas: Strategy Consulting, Management Consulting, Operations Consulting, HR Consulting, Financial Advisory, and IT Consulting.

What being a consultant actually involves:-

- Identifying the core problem or opportunity a client is facing,
- Gathering and interpreting relevant data to fully understand the situation and shape a recommended solution, and
- Guiding the rollout of that solution across the organization.

FRAMEWORKS

Frameworks give consultants a structured, repeatable way to tackle problems, reducing guesswork and keeping the analysis objective. By splitting a complicated issue into smaller building blocks, they make it far easier for both consultants and clients to see what's really driving the problem and how to act on it.

MECE Principle: A way to structure problems so that each part is Mutually Exclusive, Collectively Exhaustive, meaning no overlap between categories and nothing left out, ensuring a clean, complete breakdown of an issue.

SWOT Analysis: A framework that evaluates a business or decision across four dimensions: Strengths, Weaknesses, Opportunities, and Threats, helping balance internal capabilities against external conditions.

PESTEL Principle: An analysis of the broader external environment a business operates in, covering Political, Economic, Social, Technological, Environmental, and Legal factors that could impact strategy.

DATA SCIENCE

Data Science is an interdisciplinary field that combines statistics, programming, and domain knowledge to extract meaningful insights from data. It helps organizations identify patterns, make informed decisions, and predict future outcomes. The typical data science workflow includes data collection, cleaning, analysis, visualization, model building, and deployment.

Key Python Libraries

NumPy: Efficient numerical computations and array operations.
Pandas: Data manipulation and analysis using DataFrames.
Matplotlib: Creating graphs, charts, and visualizations.
Scikit-learn: Machine learning algorithms and model evaluation tools.
TensorFlow/PyTorch: Building and training deep learning models.

MACHINE LEARNING

Machine Learning is a branch of Artificial Intelligence that enables computers to learn from data and improve performance without explicit programming. Models are trained using historical data to recognize patterns and make predictions.

LINEAR REGRESSION :
A supervised learning algorithm used to predict continuous values by fitting the best possible line through data points. Common applications include sales forecasting and house price prediction.

LOGISTIC REGRESSION :
Used for classification tasks where the output belongs to predefined categories, such as spam detection or disease diagnosis.

MEAN SQUARED ERROR :
A performance metric that measures the average squared difference between actual and predicted values. Lower MSE values indicate better model accuracy and predictive performance.

OSAT & ATMP POLICY GUIDE

Defining OSAT & ATMP Criteria

What is an OSAT?

Outsourced Semiconductor Assembly and Test providers represent the essential "back-end" of chip manufacturing. They turn delicate, fabricated silicon wafers into durable, sellable semiconductor units. OSATs are volume-driven businesses. They command thin, stable margins—typically capturing 8% to 15% of the total integrated circuit cost. Commercial Scale Requirement: Minimum eligible investment threshold of $6.5 Million (~₹50 Crore) is required to qualify.

Qualifying as ATMP

A facility must perform all four sequential steps in India to officially qualify as an ATMP unit:

A for Assembly : Dicing silicon wafers into individual dies and wire-bonding them to substrates.

T for Testing : Rigorous electrical and functional verification to filter out defective units.

M for Marking : Laser-etching model numbers, tracking codes, and branding on each package.

P for Packaging : Encapsulating the fragile assembly in a protective plastic or ceramic mold.

SCHEMES, SUBSIDIES & SUPPORT BASIS

50%

CENTRAL CAPEX SUBSIDY

Core Incentive Schemes Offered

Modified ATMP/OSAT Scheme: The flagship program under the India Semiconductor Mission (ISM) offering a uniform 50% central capex subsidy on a pari-passu (equal-footing) basis. State

Semiconductor Policies: Stackable regional top-up schemes (e.g., Gujarat, Uttar Pradesh, Assam) that provide 20% to 30% additional capex support, alongside land subsidies and utility concessions.

ISM 1.0 to ISM 2.0 Shift: Transitioning from capacity-based subsidies to supply chain ecosystem support, focusing on local specialty chemicals, materials, and back-end integration.

LET'S TAKE A BREAK

INDIAN SEMICONDUCTOR MISSION (ISM)

What Is ISM?

Launched December 2021, under MeitY, with a ₹76,000 crore (~$10B) outlay — India's flagship push to build a domestic semiconductor ecosystem.

Schemes Under ISM:

SPECS/ATMP Scheme — up to 30% capex support for OSAT/packaging units (this is the scheme directly relevant to our client)

Semiconductor Fabs Scheme — up to 50% subsidy on project cost for fab projects

DLI Scheme — supports chip design startups across design-development-deployment stages

Live Proof Points

- Your paragraph textTata ATMP (Assam) — 48M chips/day capacity
- Micron ATMP (Gujarat) — ~$2.75B investment
- HCL-Foxconn (Jewar)
- India hosts around 20% of global semiconductor design engineers, giving it a structural edge in design/IP even as manufacturing scales

State-Level Stacking

Gujarat, Tamil Nadu, Telangana, Karnataka offer additional capex/power/land incentives layered on top of central ISM support — effective subsidy can exceed the headline 30%

What is ISM 2.0?

- ISM 2.0 builds on the progress of the first phase and aims to create a self-reliant, end-to-end semiconductor ecosystem in India
- Marks a shift from policy formulation and capacity creation to consolidation, technological depth, and global integration
- Focus areas: producing semiconductor equipment and materials in India, designing full stack Indian semiconductor IP, and fortifying domestic and global supply chains

Strategic Importance of ISM 2.0

- Global supply chain realignment under China+1 and geopolitical risk is opening space for India as a trusted alternative semiconductor base
- Risk flagged publicly: execution discipline matters — success of the first 10 projects will be the real test of whether funds translate into output, not just announcements

Role in our project

ISM (via SPECS/ATMP) is the single largest lever in our IRR model — modelled as a 30%-of-capex Year 1 inflow, stress-tested for delay/cap/withdrawal scenarios (Slide 6 of our financial model)

FINANCIAL MODELLING

A quantitative assessment of the total market demand for a product or service, usually measured in terms of revenue, number of customers, or units sold over a specific period and geographic area.

MEASURABLE RESULTS

COST-EFFICIENCY

SCALABILITY

BEST PRACTICES FOR PERFORMANCE MARKETING

01 SET CLEAR OBJECTIVES

02 CONTINUOUS TESTING

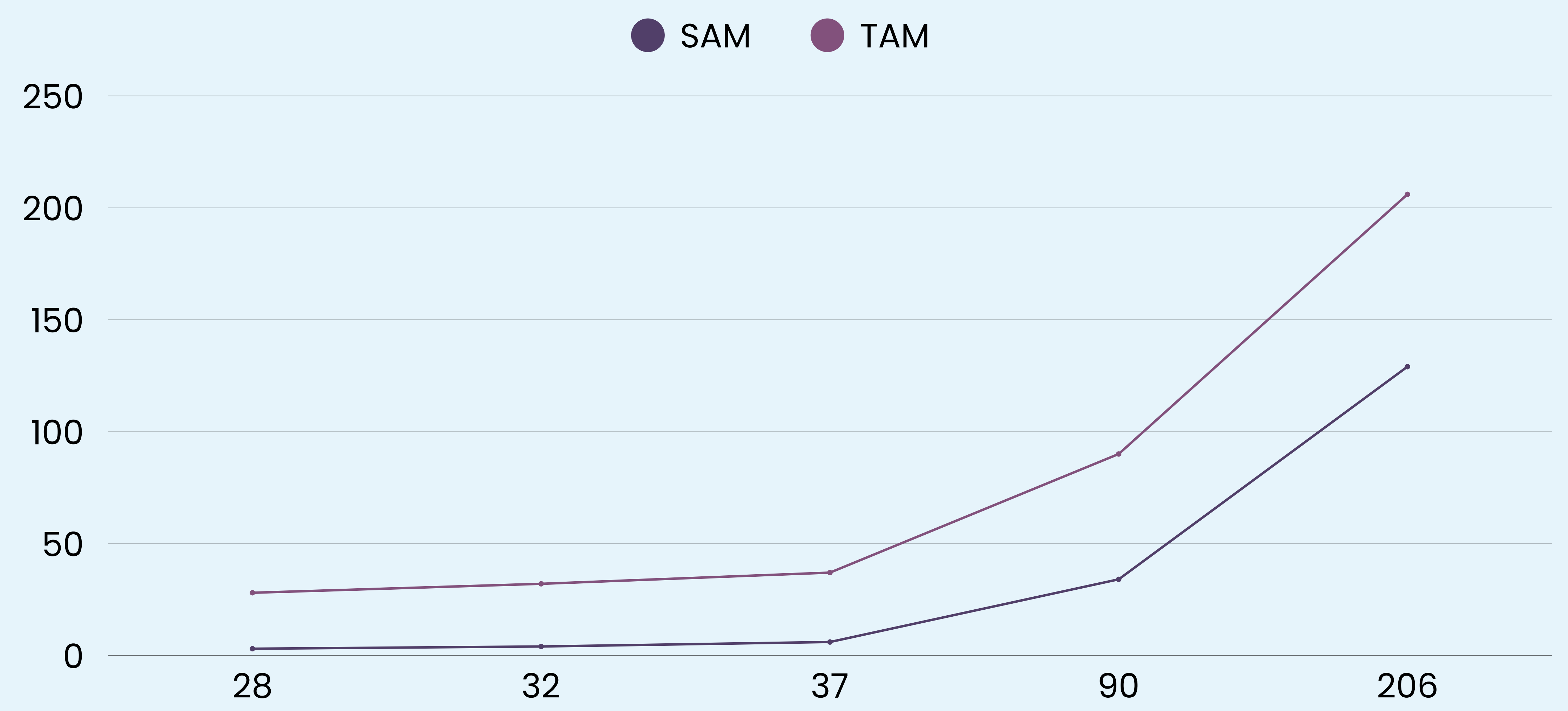
03 TARGET THE RIGHT AUDIENCE

04 SOCIAL MEDIA ADVERTISING

To get the most out of performance marketing, it’s crucial to set clear objectives from the outset. Defining your desired outcomes ensures that your campaigns remain focused and results-oriented. Additionally, continuous testing, audience targeting, and strategic budget allocation are essential best practices for optimizing your campaign’s performance and driving better results.

MARKET SIZING

A quantitative assessment of the total market demand for a product or service, usually measured in terms of revenue, number of customers, or units sold over a specific period and geographic area.



- Indian semiconductor demand expected to grow from $28B (FY23) to $206B (FY35)
- Projected 19% CAGR, significantly higher than the global semiconductor growth rate
- Current 90–95% import dependence creates a major domestic manufacturing opportunity
- India aims to build a $120–150B semiconductor value chain by 2035
- Focus sectors: AI, EVs, 5G/6G, Data Centres, Defence

STATES IMPLEMENTING OSAT AND THEIR CHALLENGES

Gujarat (Sanand-Dholera)	Live	Three OSAT/ATMP lines now in commercial production (Micron, Kaynes, CG Power) - but simultaneous ramp-up is straining grid power and ultra-pure water supply across the cluster.
Assam (Jagiroad)	Building	Tata's Rs 27,000 Cr OSAT facility required a dedicated Rs 111 Cr water-supply project and barge transport via the Brahmaputra for heavy equipment; local process-engineering talent base still nascent
Karnataka (KWIN City, Bengaluru)	Stalled	Lost a 2022 anchor MoU with a semiconductor consortium despite a strong design-talent base; the planned 200-acre KWIN semiconductor park remains pre-construction, with no assembly/packaging ecosystem yet in place
Uttar Pradesh (Jewar/YEIDA)	Lagging	Offers the most generous package among states full capex reimbursement - yet has struggled to convert incentives into an anchor OSAT investment; HCL-Foxconn's display-fab JV remains the closest proof point

Pattern: states that pair strong incentives with fast land and utility delivery (Gujarat, Assam) are converting policy into production; those without execution speed (Karnataka, UP) are not.

INDIA SEMICONDUCTOR MISSION (ISM)

ISM IN INDIA'S UNION BUDGET

ISM Total Outlay
(Approved 2021)

₹76,000 Cr
≈ $10 Billion

Union Budget
FY 2025–26

₹6,903 Cr
MeitY-ISM Allocation

ISM 2.0 FY26–27
Announced

₹1,000 Cr
R&D + Talent + Mfg

HOW ISM APPEARS IN THE BUDGET

Ministry	MeitY (Ministry of Electronics & IT)
Budget Head	Major Head 2852 — Industries (Electronics)
Sub-Head	India Semiconductor Mission (ISM)
Capex Subsidy Route	SPECS / ATMP Scheme — 50% of project cost
Disbursement	Phased reimbursement against verified milestones

WHAT THIS MEANS FOR OUR CLIENT

- 30% capex subsidy under SPECS/ATMP (50% central + state top-up) — modelled as Year 1 inflow in our IRR
- Budget allocation is annual; ISM 2.0's ₹1,000 Cr signals continued political commitment through FY27
- Milestone-linked disbursement protects govt; client must plan 18–24 month cash-bridge before first reimbursement
- Subsidy phase-down risk: if ISM 2.0 budget is cut, effective support can fall below 30%

INDIA SEMICONDUCTOR MISSION (ISM)

BUDGET TIMELINE & SUBSIDY STRESS-TEST

ISM BUDGET ALLOCATION HISTORY

SUBSIDY SCENARIO ANALYSIS

Base Case	50% (SPECS + State)	**~18%**
Delay Case	50% but 18-mo delay	**~14%**
Phase-Down	Falls to 30%	**~10%**

Peak allocation
Micron + Tata

ISM 2.0 R&D
focus

COMPETITIVE LANDSCAPE BEYOND INDIA

Country	Major Players	STRENGTHS	Weakness
Taiwan	ASE/SPIL	World’s Largest packaging ecosystem	High labor cost
China	JCET/Tongfu	Massive government support	US export restrictions
Vietnam	Amkor Vietnam	Cheap labor, growing FDI	Smaller ecosystem
India	Tata Electronics	Huge incentives & low labor cost	Supply chain still developing

Global Market Share

Approximate OSAT market share:

Taiwan → 50–55%

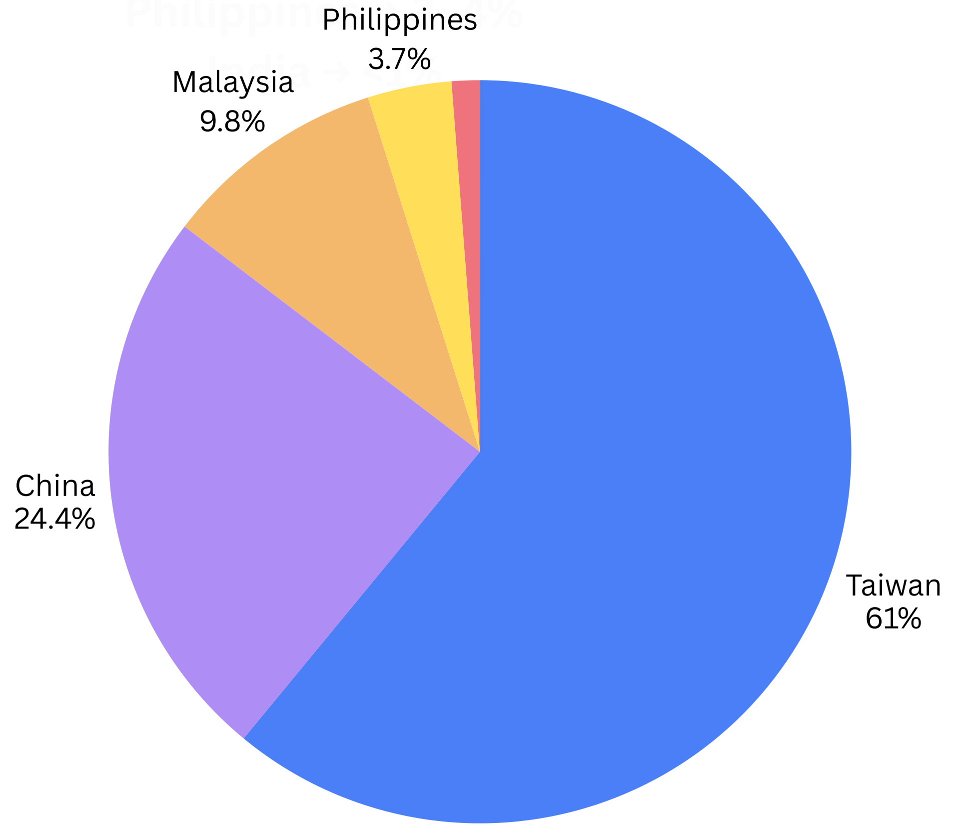
China → 20–25%

Malaysia → 7–8%

Singapore → 4–5%

Philippines → 3–4%

India → <1%



India’s Position

Advantages:

- **Very low labor costs**
- **Government subsidy up to 50% of project cost under semiconductor initiatives**
- **Large electronics manufacturing market**

SUPPLY CHAIN DEPENDENCIES

What is Supply Chain Definition:

Supply chain is the network involved in sourcing raw materials, manufacturing, assembly, testing, logistics, and delivery of products to customers.

Unlike conventional manufacturing, semiconductor production requires precision logistics, uninterrupted utilities, and highly specialized materials, making supply chain efficiency a major competitive advantage

FOR SEMICONDUCTOR OSAT

1. ***SILICON WAFER***
2. ***OSAT***
3. ***Fabrication Plant (Fab)***
4. ***Electronics Manufacturer***
5. ***Customer***

: **Major Dependencies**

1. Imported Machinery

India imports:

- Packaging machines
- Bonding machines
- Testing equipment
- Inspection systems

Mostly from:

- Japan

2. Raw Materials

OSAT requires:

- Lead frames
- Substrates
- Gold wire
- Copper wire
- Epoxy molding compounds
- Solder balls

3. Skilled Workforce

Requires:

- *Semiconductor engineers*
- *Packaging specialists*
- *Process engineers*
- *Reliability testing experts*

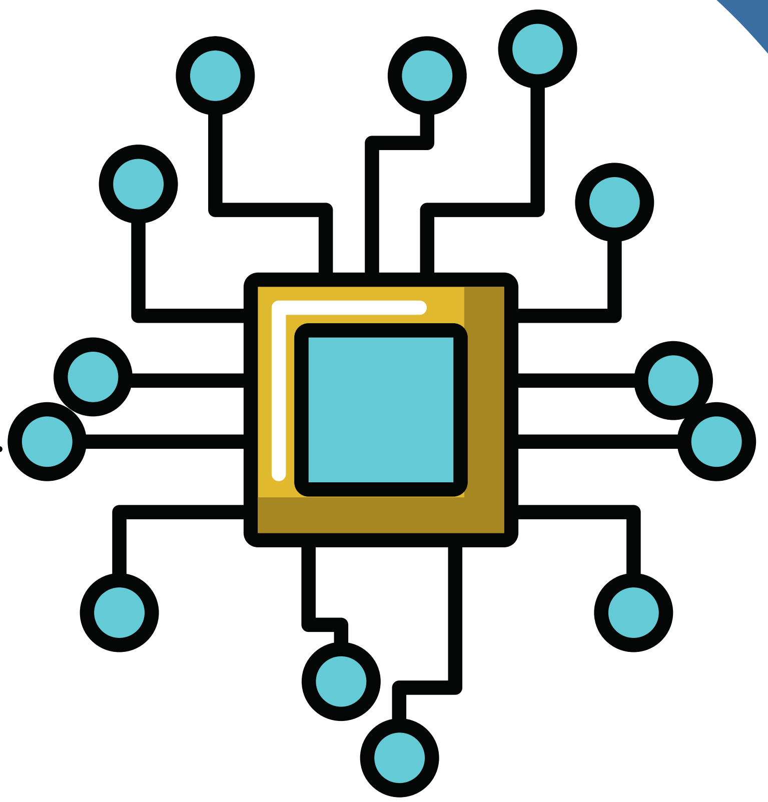
India currently has a shortage.

4. Logistics

Semiconductor supply chains require:

- *Fast airports*
- *Stable electricity*
- *Clean rooms*
- *Temperature-controlled logistics*
- *Efficient ports*

Delays increase costs significantly.



MAJOR RISKSt

DEPENDENCY	Challenge	Business Impact
IMPORTED CHIPS	GLOBAL SHORTAGES	PRODUCTION DELAYS
IMPORTED MACHINERY	HIGH CAPITAL COST	*EXPENSIVE EXPANSION*
IMPORTED MATERIALS	CURRENCY FLUCTUATIONS	INCREASED OPERATING COST
FOREIGN TECHNOLOGY	EXPORT RESTRICTIONS	TECHNOLOGY ACCESS ISSUES
Skilled Manpower	TALENT SHORTAGE	LOWER PRODUCTIVITY

India's Current Challenges

- ***Weak domestic supplier ecosystem***
- ***Heavy reliance on imports***
- ***Limited advanced packaging expertise***
- ***High dependence on foreign technology***
- ***Lack of indigenous semiconductor equipment manufacturing***
- ***Supply chain vulnerability to geopolitical events***

Strategic Recommendations

Short Term (1–3 Years)

Attract global OSAT companies

Import advanced equipment

Develop semiconductor training programs

Long Term (7–10 Years)

- ***Indigenous semiconductor equipment manufacturing***
- ***Advanced packaging R&D***
- ***Local testing equipment ecosystem***
- ***Reduced import dependence***

INDIA SEMICONDUCTOR MISSION (ISM) — OSAT ENTRY

INVESTOR RISK ASSESSMENT

Top 5 Investor Risks

1. Subsidy Phase-Down
ISM 2.0's FY26-27 budget is just ₹1,000 Cr — support may fall below 50%.

2. Ecosystem Import Dependence
Equipment, chemicals & substrates still imported — no local depth yet.

3. Output Still Unproven
₹1.6 lakh Cr approved (10 projects) but most remain in build-out.

4. Automotive Timeline Risk
Auto-grade OSAT capacity remains 18-24 months away per operators.

5. Competitive Catch-Up
Malaysia training 60K engineers by 2030; Vietnam funding 50K.

Live Risk Signals

- Industry ops report daily shortages of materials, equipment & trained engineers
- ISM 1.0 built policy credibility, but limited actual manufacturing output so far
- Govt itself is shifting from capex subsidy to ecosystem-building — a tell on what's missing

COMPETITIVE BENCHMARKING

CAN INDIA COMPETE WITH FOREIGN HUBS?

Where India Competes Today

Stacked Subsidy Depth

30% central (SPECS/ATMP) + ~20% Gujarat = more generous than most peer capex schemes.

Price Parity Claims

Indian OSATs now benchmark against Malaysia, claiming cost parity in legacy/mature packaging.

Domestic Demand + Talent

Large captive market + existing design-engineer base gives a structural starting edge.

Where India Still Trails

Ecosystem Maturity Gap

Malaysia remains the closest comparable OSAT model — denser supply chain, lower execution risk.

Import Dependence

Equipment, materials & tools still imported — limits true cost/IP sovereignty.

Peers Investing Too

Malaysia & Vietnam both scaling talent pipelines aggressively — the gap isn't static.

VERDICT: Strong enough to win the capital allocation decision today — not yet strong enough to claim structural superiority over Malaysia.

BOTTOM LINE

The subsidy gets India into the conversation. Ecosystem depth decides whether it stays a price-taker or becomes a durable #2/#3 hub once incentives taper.

Subsidy is sufficient to win Day-1 capital allocation: 30% central + ~20% state stacking is genuinely competitive on paper, and now also in early legacy-packaging cost benchmarks vs Malaysia.

Ecosystem depth is the real competitive gap — not subsidy size: Equipment, materials, and skilled-technician supply chains remain import-dependent. This is what ISM 2.0 itself is now trying to fix.

India's structural edge is demand + talent + geopolitics, not headline incentives: Domestic market size, existing design-engineer base, and China+1 tailwinds are the durable differentiators — not the subsidy percentage alone.

THANK YOU